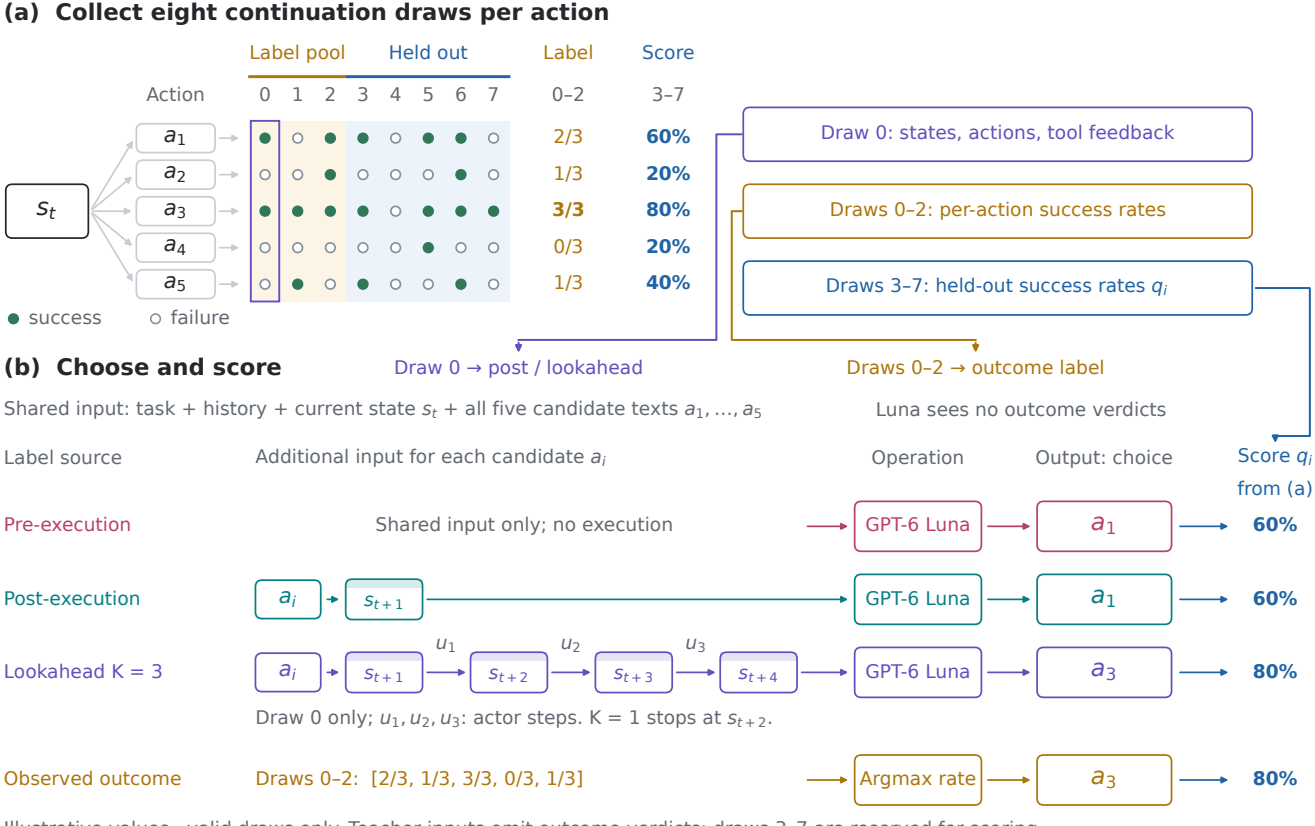
(a) Collect eight continuation draws per action
Label pool
Held out
Label
Score
Action
0 1 2 3 4 5 6 7
0–2
3–7
a_1 2/3 60%
a_2 1/3 20%
a_3 3/3 80%
a_4 0/3 20%
a_5 1/3 40%
s_t
success
failure
Draw 0: states, actions, tool feedback
Draws 0–2: per-action success rates
Draws 3–7: held-out success rates q_i
Draw 0 → post / lookahead
Draws 0–2 → outcome label
(b) Choose and score
Shared input: task + history + current state s_t + all five candidate texts $a_1, \ldots, a_5$
Luna sees no outcome verdicts
Label source
Additional input for each candidate a_i
Operation
Output: choice
Score q_i from (a)
Pre-execution
Shared input only; no execution
GPT-6 Luna
a_1
60%
Post-execution
a_i
s_{t+1}
GPT-6 Luna
a_1
60%
Lookahead K = 3
a_i
s_{t+1}
u_1
s_{t+2}
u_2
s_{t+3}
u_3
s_{t+4}
GPT-6 Luna
a_3
80%
Draw 0 only; u_1, u_2, u_3: actor steps. K = 1 stops at s_{t+2}.
Observed outcome
Draws 0–2: [2/3, 1/3, 3/3, 0/3, 1/3]
Argmax rate
a_3
80%
Illustrative values · valid draws only. Teacher inputs omit outcome verdicts; draws 3–7 are reserved for scoring.